TortuBot – ESP32 Educational Mobile Robot Platform

Project Description

TortuBot is an open-source educational mobile robotics platform built around the **ESP32 microcontroller**, designed to provide a practical environment for learning, experimentation, and rapid prototyping of autonomous and remote-controlled robots. The platform integrates essential robotics components, including DC motor control, wheel encoders, infrared sensors, and communication interfaces, enabling users to explore fundamental robotics concepts through real-world applications.

The ESP32 serves as the central controller, managing sensor data acquisition, motor control, and wireless communication. With built-in **Wi-Fi connectivity**, TortuBot supports HTTP-based communication, remote monitoring, and cloud integration, allowing users to develop Internet of Things (IoT) robotics applications. Its modular hardware and open-source software architecture make it easy to customize, expand, and integrate additional sensors and peripherals.

The project emphasizes hands-on learning, embedded systems programming, robotics algorithms, and IoT development, making it an ideal platform for students, educators, researchers, and robotics enthusiasts.

Hardware Components

Component	Quantity	Purpose
ESP32 Development Board	1	Main microcontroller for robot control and communication
DC Geared Motors	2	Robot locomotion
Motor Driver Module	1	Controls motor speed and direction
Wheel Encoders	2	Measures wheel rotation for odometry
Infrared (IR) Sensors	2–5	Obstacle detection and line-following
Robot Chassis	1	Mechanical structure
Wheels	2	Robot movement
Caster Wheel	1	Stability and balance
Rechargeable Battery	1	Portable power supply
Power Management Circuit	1	Voltage regulation and distribution
Wi-Fi Module (Integrated in ESP32)	1	Wireless communication and IoT connectivity

Component	Quantity	Purpose
Jumper Wires & Connectors	As Required	Electrical connections

Key Features

- ESP32-based mobile robot platform
- Differential drive motor control
- Wheel encoder support for odometry
- Infrared sensor integration for obstacle detection and line following
- Built-in Wi-Fi communication
- HTTP server/client functionality
- Cloud and IoT integration support
- Modular hardware and software architecture
- Open-source firmware and hardware design
- Ideal for robotics education, experimentation, and rapid prototyping

Applications

- Robotics education and training
- Mobile robot prototyping
- Autonomous navigation research
- Line-following robots
- Obstacle avoidance systems
- IoT-enabled robotic applications
- Embedded systems learning
- STEM and engineering laboratory projects